



RESEARCH & INDUSTRY EXPERIENCE

Senior Data Scientist

Vaalia Health

📅 March 2025 – Present

📍 Espoo, Finland

- Developed Transformer-based multi-target forecasting models for longitudinal healthcare data, alongside regularized baselines, robust losses, and calibration evaluation
- Designed causal-inference and counterfactual-simulation methods for intervention-effect estimation, including autoregressive baselines, stratified analysis, and documented DAG assumptions
- Productionized reproducible Databricks workflows with PySpark, PyTorch Lightning, Hydra, and MLflow, covering cohort construction, cross-validation, model lineage, and automated reporting

Lead Researcher, Aalto–Nokia Collaboration

Aalto University & Nokia

📅 July 2024 – December 2024

📍 Espoo, Finland

- Led Aalto’s research workstream on hardware-aware combinatorial optimization for emerging 6G systems
- Developed an analog–digital Ising machine model for multi-user MIMO detection, translating memristor constraints into algorithm design

Ph.D. Researcher

Aalto University — Machine Learning for Health Group

📅 August 2019 – December 2024

📍 Espoo, Finland

- Developed coupled hidden Markov models to infer latent interactions in multivariate longitudinal health data and analyze individualized treatment response
- Introduced mixture-of-CHMM models that discover heterogeneous response subgroups directly from data
- Modeled glucose–insulin dynamics with multi-output hierarchical Gaussian processes, producing uncertainty-aware personalized forecasts
- Developed sparse Kalman filtering methods that scale nonlinear state estimation to high-dimensional physical systems

Data Science Consultant

Boğaziçi University & BKM — Interbank Card Center

📅 June 2018 – June 2019

📍 Istanbul, Turkey

- Delivered a paid university–industry consulting engagement, building Spark-based machine-learning pipelines for large-scale credit-card transaction data
- Developed interpretable fraud-detection and merchant-risk models adopted in BKM’s production analytics

PROFILE

Research scientist and PhD developing probabilistic and causal machine-learning methods for longitudinal healthcare data and dynamical systems. Combines methodological research with experience translating uncertainty-aware models into real-world decision support.

EDUCATION

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Ph.D. in Computer Science

Aalto University

📅 2019 – 2024

Thesis: Flexible Bayesian latent variable modeling of interacting processes in time-series healthcare

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M.Sc. in Computational Science and Engineering

Boğaziçi University

📅 2017 – 2019

Thesis: Anomaly Detection in Time Series

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B.Sc. in Electrical and Electronics Engineering

Boğaziçi University

📅 2010 – 2016

RESEARCH AREAS

- Probabilistic & Bayesian modeling
- Causal inference & counterfactuals
- Longitudinal healthcare data
- Forecasting & state-space models
- Uncertainty & calibration
- Sequential decision systems

RECOGNITION

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Top 0.006% nationally

Ranked 112th among approximately 2 million participants in Turkey’s university entrance examination.

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BUALERT engineering team

Designed car electronics, ECU, and energy-management systems for an alternative-energy racing team.

EARLIER EXPERIENCE

Data Scientist Digital Helix – DeepC

June 2018 – December 2018 Istanbul, Turkey

- Developed supervised and unsupervised deep-learning workflows for brain-lesion detection in MRI scans
- Worked across image segmentation, classification, and feature regression on high-dimensional medical images

M.Sc. Researcher Boğaziçi University & Borusan – R&D

February 2017 – April 2018 Istanbul, Turkey

- Developed unsupervised anomaly-detection models for industrial wind-turbine sensor streams
- Built Bayesian and deep time-series models for real-time detection in noisy multivariate signals

PEER-REVIEWED PUBLICATIONS

Journal Articles

- Poyraz, Onur, Sini Heinonen, et al. (2025). “Computational modeling enables individual assessment of postprandial glucose and insulin responses after bariatric surgery”. In: *Communications Medicine* 5.1, p. 434.
- Poyraz, Onur, Mohamad R. A. Sater, et al. (2022). “Modelling MRSA decolonization: interactions between body sites and the impact of site-specific clearance”. In: *Journal of the Royal Society Interface* 19.191, p. 20210916. DOI: 10.1098/rsif.2021.0916.

Conference Proceedings

- Haitsiukevich, Katsiaryna et al. (2024). “Diffusion Models as Probabilistic Neural Operators for Recovering Unobserved States of Dynamical Systems”. In: *2024 IEEE 34th International Workshop on Machine Learning for Signal Processing (MLSP)*, pp. 1–6. DOI: 10.1109/MLSP58920.2024.10734762.
- Poyraz, Onur and Pekka Marttinen (2023). “Mixture of Coupled HMMs for Robust Modeling of Multivariate Healthcare Time Series”. In: *Proceedings of the 3rd Machine Learning for Health Symposium*. Vol. 225. PMLR, pp. 461–479.
- Poyraz, Onur, Cem Say, and Ali Taylan Cemgil (2020). “Merchant Risk Scoring”. In: *Signal Processing and Communications Applications Conference (SIU), 2020 28th*. IEEE, pp. 1–4.
- Poyraz, Onur, Burak Kurutmaz, et al. (2018). “Anomaly Detection on Wind Turbines”. In: *Signal Processing and Communications Applications Conference (SIU), 2018 26th*. IEEE, pp. 1–4.

ACADEMIC SERVICE & COMMUNITY

- Scientific Symposium, by Students for Students – Volunteer, May 2021, Espoo
- Machine Learning at the Bosphorus İsmail Arı Summer School – Volunteer, July 2018, İstanbul

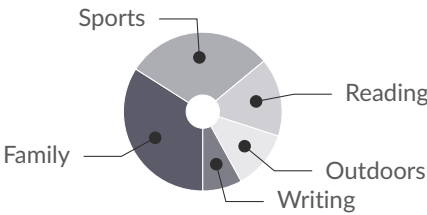
METHODS & TOOLS

- Programming**
Python, R, SQL, C/C++, Julia, MATLAB
- Modeling**
PyTorch, PyTorch Lightning, Stan, TensorFlow
- Data systems**
Databricks, PySpark/Spark
- Reproducibility**
MLflow, Hydra/OmegaConf, pytest, GitHub Actions

LANGUAGES

- Turkish**
Native
- English**
Professional fluency
- Finnish**
Intermediate

PERSONAL INTERESTS



REFERENCES

- Assoc. Prof. Pekka Marttinen**
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Aalto University
- Prof. Ali Taylan Cemgil**
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Department of Computer Engineering
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